ChatGPT said:

Yes — this is the right next step.

Your **industry risk repo should be restructured into a reusable reference-layer project**, especially for the property-backed PD, LGD, and EL work. For property-backed lending, the most useful public Australian inputs are:

- **ABS Building Approvals** for pipeline / regional construction activity,
- **ABS Building Activity** for commencements, completions and related construction-cycle signals,
- **ABS Lending Indicators** for housing finance activity,
- **RBA housing arrears / Financial Stability Review material** for the arrears environment and housing stress context, and
- **APRA ADI / property exposure statistics** for system-level banking and housing exposure context. Australian Bureau of Statistics+3

Below is the clean way to rebuild the repo.

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1. What this repo should become

Do **not** position this repo as a PD model or an EL model.

Position it as:

Australian Property & Industry Risk Reference Layer

Its job is to produce **reusable reference tables** that other projects can consume:

- property-backed PD repo
- LGD repo
- EL engine repo

So the logic becomes:

industry risk repo → reference tables → used by PD / LGD / EL

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2. What you should build inside this repo

Build **four outputs** for property-backed lending:

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1. **region risk bands**
2. **market softness / construction-cycle bands**
3. **base arrears environment**
4. **simple downturn overlays**

These should be **reference outputs**, not final loan-level model outputs.

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3. How to restructure the repo

Use a cleaner structure like this:

```
industry-risk-reference-layer/
│
├── README.md
├── requirements.txt
├── .gitignore
│
├── data/
│  ├── raw/
│  │  ├── abs/
│  │  ├── apra/
│  │  ├── rba/
│  │  └── manual/
│  ├── processed/
│  │  ├── property_reference/
│  │  └── industry_reference/
│  └── output/
│  ├── region_risk/
│  ├── property_cycle/
│  ├── arrears_environment/
│  └── downturn_overlays/
│
├── notebooks/
│  ├── 01_project_overview_and_data_map.ipynb
│  ├── 02_abs_property_pipeline_and_cycle.ipynb
│  ├── 03_apra_rba_arrears_environment.ipynb
```

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```
│ ├─ 04_region_risk_banding.ipynb
│ ├─ 05_property_cycle_banding.ipynb
│ ├─ 06_downturn_overlay_design.ipynb
│ └─ 07_final_reference_outputs.ipynb
│
├─ src/
│ ├─ config.py
│ ├─ data_loader_abs.py
│ ├─ data_loader_apra.py
│ ├─ data_loader_rba.py
│ ├─ region_risk.py
│ ├─ property_cycle.py
│ ├─ arrears_environment.py
│ ├─ downturn_overlay.py
│ └─ utils.py
│
└─ docs/
├─ project_overview.md
├─ methodology_region_risk.md
├─ methodology_property_cycle.md
├─ methodology_arrears_environment.md
├─ methodology_downturn_overlays.md
└─ limitations_and_assumptions.md
```

---

4. What each new output should do

A. Region risk bands

This table should summarise **regional property / construction risk**.

Suggested output

`data/output/region_risk/region_risk_table.csv`

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Suggested columns

> region
> state
> region_group
> building_approvals_trend
> building_activity_trend
> housing_finance_trend
> region_risk_score
> region_risk_band
> as_of_date
> source_note

How to build it

Use:

- ABS Building Approvals
- ABS Building Activity
- ABS Lending Indicators

These releases give you market activity, approvals, construction timing, completions, and housing finance signals you can convert into simple regional bands. Australian Bureau of Statistics+1

Banding logic example

- low risk = stronger approvals + steadier activity + solid finance demand
- medium risk = mixed signals
- high risk = falling approvals + weak activity + softer finance demand

This should be a **reference table**, not a default model.

B. Market softness / construction-cycle bands

This table should describe where each region or property segment sits in the cycle.

Suggested output

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```
data/output/property_cycle/property_cycle_table.csv
```

Suggested columns

> region
> property_segment
> approvals_change_pct
> commencements_signal
> completions_signal
> cycle_stage
> market_softness_score
> market_softness_band
> as_of_date
> source_note

How to build it

Mainly use:

- ABS Building Approvals
- ABS Building Activity

ABS states that Building Activity data includes average commencement times, completion times, and abandonment rates in its data downloads, which is useful for development and construction-cycle framing. [Reserve Bank of Australia]

Cycle stages

Use simple labels:

- growth
- neutral
- slowing
- downturn

This output is especially useful for:

- property-backed PD overlays
- development-lending PD

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- LGD downturn haircuts

C. Base arrears environment

This should capture the current housing stress / arrears background in Australia.

Suggested output

`data/output/arrears_environment/base_arrears_environment.csv`

Suggested columns

> as_of_date
>
> arrears_environment_level
>
> arrears_trend
>
> macro_housing_risk_band
>
> macro_housing_risk_score
>
> notes
>
> source_note

How to build it

Use:

- RBA Financial Stability Review / housing arrears commentary
- APRA property exposure / banking statistics where useful

The RBA's March 2026 Financial Stability Review says housing loan arrears are low and have continued to decline, returning to around pre-pandemic levels, while still discussing higher vulnerability among some more leveraged and lower-income borrowers. [Reserve Bank of Australia+1]

Why this matters

This gives you a **macro arrears baseline** to use later in:

- property PD overlays
- EL stress scenarios

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- discussion of current lending conditions

Do not try to make this too detailed. It is a **macro reference flag**, not a borrower feature table.

D. Simple downturn overlays

This should be a scenario table that other repos can read directly.

Suggested output

`data/output/downturn_overlays/property_downturn_overlays.csv`

Suggested columns

> scenario
> pd_multiplier
> lgd_multiplier
> ccf_multiplier
> property_value_haircut
> notes
> as_of_date

Example rows

> base 1.00 1.00 1.00 0.00
> mild 1.20 1.10 1.05 0.05
> moderate 1.50 1.20 1.10 0.10
> severe 2.00 1.30 1.20 0.20

How to design it

Anchor the reasoning using:

- current arrears environment from RBA
- construction-cycle weakness from ABS
- system banking/property context from APRA

Keep the numbers clearly labelled as **illustrative portfolio assumptions**, not calibrated regulatory stress parameters. [Reserve Bank of Australia+2]

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5. Notebook-by-notebook build plan

Notebook 1 — project overview and data map

Purpose:

- explain what the repo is
- list source datasets
- define target outputs
- show how downstream projects will use the outputs

Include:

- repo purpose
- source map
- file structure
- output architecture

Notebook 2 — ABS property pipeline and cycle

Purpose:

- prepare Building Approvals, Building Activity, and Lending Indicators inputs

Use:

- ABS Building Approvals
- ABS Building Activity
- ABS Lending Indicators Australian Bureau of Statistics+1

Outputs:

- cleaned regional property activity tables
- processed trend variables

Notebook 3 — APRA/RBA arrears environment

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Purpose:

- summarise the current national housing arrears and macro housing-risk environment

Use:

- RBA FSR arrears commentary
- APRA system/property statistics as context [Reserve Bank of Australia+2]

Outputs:

- base arrears environment table
- short narrative note for README/docs

Notebook 4 — region risk banding

Purpose:

- build region risk scores and bands from processed public data

Outputs:

- `region_risk_table.csv`

Notebook 5 — property cycle banding

Purpose:

- convert approvals / activity / completions into market-softness and cycle-stage bands

Outputs:

- `property_cycle_table.csv`

Notebook 6 — downturn overlay design

Purpose:

- design reusable scenario multipliers for PD / LGD / CCF / property values

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Outputs:

- `property_downturn_overlays.csv`

Notebook 7 — final reference outputs

Purpose:

- validate and export all final reference tables cleanly

Outputs:

- final CSVs in `data/output/`

---

6. How to use this repo in your other projects

This is the most important design principle.

In the property-backed PD repo

Merge:

- `region_risk_table.csv`
- `property_cycle_table.csv`
- `base_arrears_environment.csv`

Then use them as:

- input features
- overlay drivers
- score / PD uplifts

Example logic:

- higher region risk band → PD uplift

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- downturn cycle stage → higher PD
- weaker arrears environment → slightly more conservative calibration

---

In the LGD repo

Use:

- region risk
- market softness
- downturn overlays

Then apply:

- higher recovery haircut
- stronger forced-sale assumptions
- higher downturn LGD

In the EL engine

Use:

- downturn overlays table directly in scenario testing

Example:

- stressed PD = pd_final × pd_multiplier
- stressed LGD = lgd_final × lgd_multiplier
- stressed EAD = ead × ccf_multiplier where relevant

---

7. How to build the public-data logic clearly

You asked specifically whether to build the “real Australian public data” layer first.

Yes — do it in this order:

Step 1

Clean the raw ABS / APRA / RBA inputs

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Step 2

Create processed regional / cycle / arrears indicators

Step 3

Band them into reusable business categories

Step 4

Export reference tables

Step 5

Consume them in PD / LGD / EL

That is much cleaner than embedding public-data logic separately into every repo.

---

8. Best-practice cleanup for GitHub

Rename the project

Use something clearer, like:

- `industry-risk-reference-layer`
- `australian-property-industry-risk-reference`
- `industry-and-property-risk-reference-layer`

I would avoid a vague title if the repo is now becoming the shared reference layer.

Rewrite the README structure

Your README should contain:

Title

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Australian Property & Industry Risk Reference Layer

Purpose

This project builds reusable risk-reference tables for property-backed lending using Australian public data. The outputs are designed to support downstream PD, LGD and Expected Loss projects.

Public data sources

- ABS Building Approvals
- ABS Building Activity
- ABS Lending Indicators
- RBA housing arrears / Financial Stability Review
- APRA banking and property exposure statistics [Australian Bureau of Statistics+3]

Main outputs

- region risk table
- property-cycle table
- base arrears environment table
- downturn overlay table

How other repos use it

Explain the downstream integration.

Limitations

- public aggregate data only
- not loan-level borrower data
- reference-layer use only
- illustrative business banding

Add docs files

Create:

- `project_overview.md`

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- `methodology_region_risk.md`
- `methodology_property_cycle.md`
- `methodology_arrears_environment.md`
- `methodology_downturn_overlays.md`
- `limitations_and_assumptions.md`

That will make the repo feel much more complete.

Standardise outputs

Use naming aligned with your other repos:

- `region_risk_table.csv`
- `property_cycle_table.csv`
- `base_arrears_environment.csv`
- `property_downturn_overlays.csv`

This consistency matters a lot once you start connecting repos.

Clean notebooks

Make sure:

- notebook outputs are cleared before commit
- final CSVs are written to output folders
- reusable logic moves to `src/`
- README explains what each notebook produces

Improve commit history

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Use staged commits such as:

- `Refactor repo into reference-layer structure`
- `Add ABS property pipeline processing`
- `Build regional property risk bands`
- `Build arrears environment reference table`
- `Add downturn overlay scenarios`
- `Rewrite README and project docs`

That will make the repo look much more professional.

---

9. What not to do

Do not:

- calculate PD inside this repo
- calculate LGD inside this repo
- create loan-level EL here
- mix borrower-level synthetic data into this project

Keep this repo as the **shared reference layer**.

---

10. Final recommendation

Build this repo as:

a clean Australian public-data reference layer for property-backed lending

with these four final outputs:

- `region_risk_table.csv`
- `property_cycle_table.csv`

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- `base_arrears_environment.csv`
- `property_downturn_overlays.csv`

Then use those outputs in:

- your property-backed PD model
- your LGD final layer
- your EL stress testing

That is the strongest architecture for your portfolio.

If you want, next I can write the **ready-to-paste README.md** for this restructured industry risk repo.

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